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Why It's So Hard to Link Extreme Weather Events to Climate Change

Attribution science examines how global warming fuels extreme weather, but making the connection is challenging



People in Kerrville, Texas, observed the Guadalupe River on Sunday after receiving alerts on potential flooding in the area. PHOTO: MARCO BELLO/REUTERS

By [Eric Niiler](#) [Follow](#) and [Matthew Dalton](#) [Follow](#)

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Did climate change make [the Texas flood](#) worse? Scientists say answering that question is tricky.

Researchers agree that climate change has made torrential downpours more frequent across the globe. But the science gets

murky when they try to determine whether any one storm—or heat wave or drought—was fueled by global warming.

To make the connection, researchers in a field called “attribution science” examine to what extent an extreme event can be linked to climate change.

When deadly floods [washed over Texas](#) last week, scientists at ClimaMeter, a consortium of experts that analyzes the role of climate change in extreme weather, compared 30 events that were similar to the Texas floods in recent years to 30 events from 1950 to 1986, when greenhouse gas concentrations in the atmosphere were lower.

They concluded, with low confidence, that global warming likely intensified the rainfall by as much as 7%.

The assessment was based on a widely accepted finding of climate science: that greenhouse gas emissions have made the atmosphere warmer and wetter—and primed to unleash moisture in more destructive outbursts.

But the group acknowledged major limitations to its study.

Remnant moisture from Tropical Storm Barry stalled over the region and repeatedly fed rainfall, making it hard to compare the weather pattern to historical data. “For this event we have low

confidence in the robustness of our approach given the available climate data,” the study said.



Helicopters flew over Ingram, Texas, ahead of a local flash flood warning on Sunday. At right, volunteers searched the banks of the Guadalupe River on Tuesday.

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Davide Faranda, a scientist at France’s National Center for Scientific Research who co-authored ClimaMeter’s analysis, said the data nonetheless suggests that climate change played a role. “The signal we observe fits well within the theoretical expectations for a warming world,” he said.

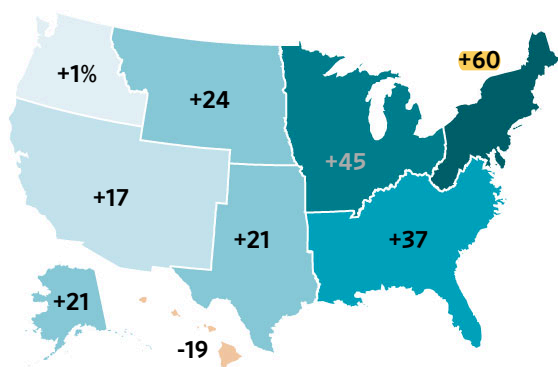
The field of attribution science is only about 20 years old, and the technique, which relies on the use of computer models that compare current and past reconstructions of the atmosphere, [is still evolving](#).

The United Nations Intergovernmental Panel on Climate Change, which assembles the most rigorous climate science in periodic reports, says that attributing short-lived weather events to climate change is difficult.

However, physics helps explain how increased warming in the atmosphere can boost rainfall.

For every 1 degree Celsius (1.8 degree Fahrenheit) increase in temperature, the air can carry about 7% more moisture—a relationship known as the Clausius-Clapeyron equation. Since the 1800s, the global atmosphere has warmed by an average 1.2 degrees Celsius (2 degrees Fahrenheit), according to a May report by the National Oceanic and Atmospheric Administration.

Change in total precipitation on heaviest 1% of days, 1958 to 2021



Source: U.S. Global Change Research Program

The agency also reported that 2024 was the warmest year on land and sea since record-keeping began in 1850, and the 10 warmest years in the 175-year record have all occurred during the past decade.

“The oceans are warmer, the air is warmer, so more evaporation is coming off the oceans, and that can feed into any storm that comes along, whether it’s a hurricane or a nor’easter or just a complex of them, like the ones that hit Texas,” said Jennifer Francis, senior scientist at the Woodwell Climate Research Center, a nonprofit institution in Falmouth, Mass.

Total rainfall in the U.S. and across the globe isn't increasing, Francis said, but the frequency of intense precipitation is on the rise, a finding backed up by peer-reviewed government and academic studies.

According to the Fifth National Climate Assessment, a congressionally mandated report [prepared by 14 federal agencies](#) that was issued in 2023, the total precipitation on the heaviest 1% of days has increased by 60% since the 1950s over the northeastern U.S., and by 21% over the region including Texas, Oklahoma and Kansas.

Academic studies have found similar results. Researchers at Dartmouth College found a 50% increase in extreme precipitation—defined as the amount falling on the 1% of wettest days—from 1996 to 2014 over the 12 northeastern states and the District of Columbia that was linked to a warmer climate, according to a 2021 study published in the journal *Weather and Climate Extremes*.



Damaged vehicles after deadly flooding in Spain's Valencia region last year; at right, volunteers cleaned flooded streets.

CESAR MANSO/AGENCE FRANCE-PRESSE/GETTY IMAGES; JORGE ZAPATA/EPA/SHUTTERSTOCK

The U.N. says that man-made climate change has likely amplified the intensity of torrential downpours in North America, Europe and Asia between 1950 and 2018, and that “heavy precipitation will generally become more frequent and more intense with additional global warming,” according to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change.

Last year, floods wreaked havoc in Central Europe over the summer, and in October [killed more than 200 people in the Valencia region of Spain](#). Spanish authorities were sharply criticized after the Valencia flooding for sending out warnings only after the rain had begun to fall—far too late to evacuate people in the path of the floodwaters.

An analysis by ClimaMeter found that climate change boosted Spanish rainfall by as much as 15%—though the group said it also has low confidence in this finding because the Valencia rains were,

like the Texas rains, unusual and difficult to compare with past events.

Justin Mankin, professor of geography at Dartmouth College, said even a small climate-change effect in a particular storm can make a big difference on the ground, for example, by pushing open a door of a home facing a flood.

“It’s not that first 10 millimeters of rainfall that matters,” he said. “Maybe it’s that 11th.”



Destruction along the Guadalupe River in Center Point, Texas, on Tuesday. PHOTO: SALGU WISSMATH FOR WSJ

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Corrections & Amplifications

Extreme rainfall has increased 21% since the 1950s in Texas,

Oklahoma and Kansas. An earlier version of this article incorrectly listed Nebraska as one of those states. (Corrected on July 10)

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